

MID-TERM PROGRESS REPORT

CHE251

A Comprehensive Comparative Analysis of the Solvay and Leblanc Processes for Sodium Carbonate Production

Date of Submission: 30 September 2025

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Abstract

This Mid-Semester Report details the problem definition and necessary inputs required to execute a comparative analysis of the historical Leblanc process and the modern Solvay process for sodium carbonate production. The core focus is on establishing the basis and critical operating conditions (conversions, purities, recycle specifications) required to perform a solvable, steady-state material and energy balance for both processes. This setup directly addresses the project objective of optimizing process design, efficiency, and environmental impact through rigorous chemical engineering principles. The report includes fully labeled flowcharts and appropriated conditions or data ensuring the problem is well-posed for subsequent computation via hand calculations, MATLAB, and the DWSIM process simulator.

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Chapter 1

Introduction and Project Overview

1.1 Background and Motivation

The production of sodium carbonate (Na_2CO_3), or soda ash, is a pillar of the industrial chemical sector. This project conducts a comprehensive analysis comparing the historically significant Leblanc process with the modern, dominant Solvay process. The shift from the Leblanc to the Solvay process represents a major evolution in chemical engineering, driven primarily by superior efficiency and sustainability metrics. The Leblanc process, developed in the late 18th century, was revolutionary but suffered from a two-stage approach: generating Na_2SO_4 from NaCl and H_2SO_4 , followed by a reduction and reaction with limestone and carbon. This method was energy-intensive and created environmentally damaging byproducts, most notably hydrogen chloride (HCl) gas and solid calcium sulfide (CaS) waste. The Solvay process, introduced in the 1860s, is characterized by its innovative use of an ammonia (NH_3) cycle and its reduced environmental footprint, now accounting for approximately three-quarters of global soda ash production. By recovering and recycling the expensive NH_3 catalyst, the process achieved economic superiority. This analysis utilizes fundamental chemical engineering principles, including material and energy balances and vapor-liquid equilibrium (VLE), to quantify the performance differences between these two landmark industrial methods.

1.2 Project Objectives

This mid-semester phase focuses on defining the problem space, specifically preparing for the detailed calculation phase. The full project objectives remain:

1. To perform a detailed steady-state material and energy balance on both processes.
2. To conduct a Degrees of Freedom analysis for a key unit operation in each process (e.g., the Solvay carbonation tower).
3. To perform a detailed economic and environmental analysis comparing raw material costs, utility consumption, and byproduct management.
4. To incorporate advanced VLE analysis in separation units to enhance modeling accuracy.

1.3 Mid-Semester Deliverable Requirements

Per the mid-semester requirements, the subsequent chapters will provide the fully annotated process flow diagrams and the detailed process specifications necessary to complete the material balance calculations, thus demonstrating the problem is fully defined and solvable (Degrees of Freedom = 0).

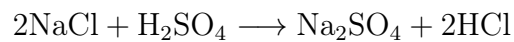
Chapter 2

Leblanc Process: Problem Definition and Specifications

This chapter provides the fully labeled flowchart and all necessary specifications for the steady-state material balance of the Leblanc process.

2.1 Key Leblanc Process Reactions

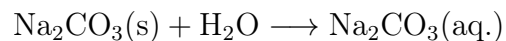
- **Reactor I:**



- **Reactor II:**



- **Leaching Tank:**



2.2 Process Flow Diagram and Stream Labeling

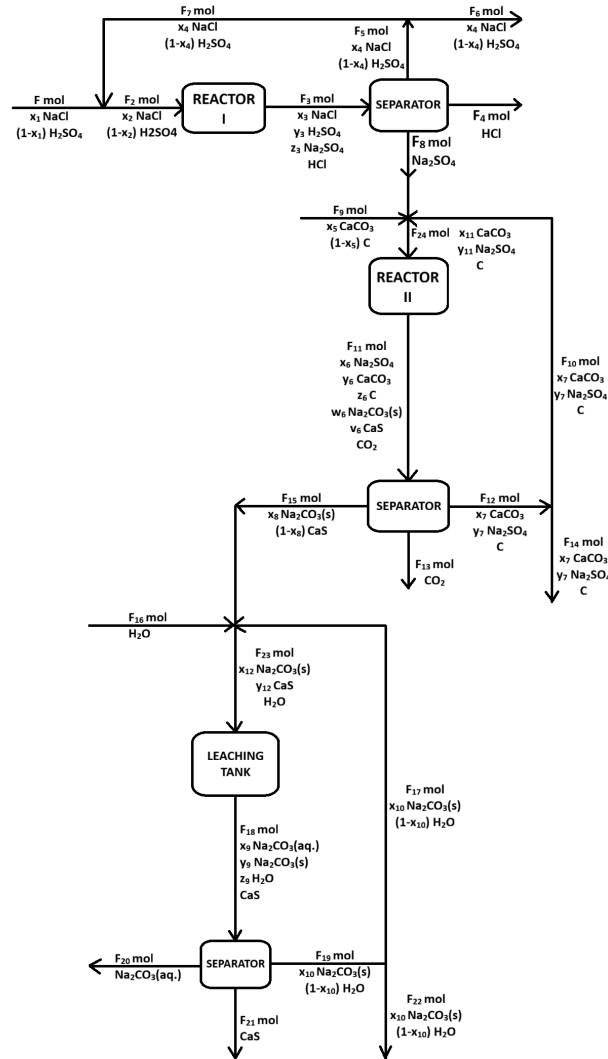


Figure 2.1: Fully Labeled Process Flow Diagram for the Leblanc Process.

Note: All the streams are mentioned with respective flow rates (F, F_i) and component mole fractions (x_i, y_i, z_i).

2.3 Basis and Given Conditions (Material Balance Specifications)

1. **Basis:** 1000 kmol/h of final dry Na_2CO_3 product. (Same as Solvay for comparison)
2. **Reactor 1 (Salt Cake Furnace) Conversion:** NaCl conversion to Na_2SO_4 is 98%.

3. **Reactor 2 (Black Ash Furnace) Conversion:** The fractional conversion of Na_2SO_4 (relative to the feed Na_2SO_4) is 85 – 95%.
4. **Leaching Tank:** Solubility of Na_2SO_4 in water at 50°C is 47.3 g per 100 g water; conversion to aqueous phase = 90 – 97%.

2.4 Operating Conditions and Unit Conversions (Provided Data)

Unit Operation	Temperature (T)	Pressure (P)	Conversion / Solubility
Reactor I	600 – 700°C	1 atm	Single-Pass Conversion (NaCl): $\sim 98\%$
Reactor II	1000 – 1100°C	1 atm	Single-Pass Conversion (Na_2SO_4): 85 – 95%
Leaching Tank	50 – 70°C	1 atm	Single-Pass Conversion (Na_2CO_3 dissolution): 90 – 97%

Chapter 3

Solvay Process: Problem Definition and Specifications

This chapter provides the fully labeled flowchart and all necessary specifications for the steady-state material balance of the Solvay process.

3.1 Process Flow Diagram and Stream Labeling

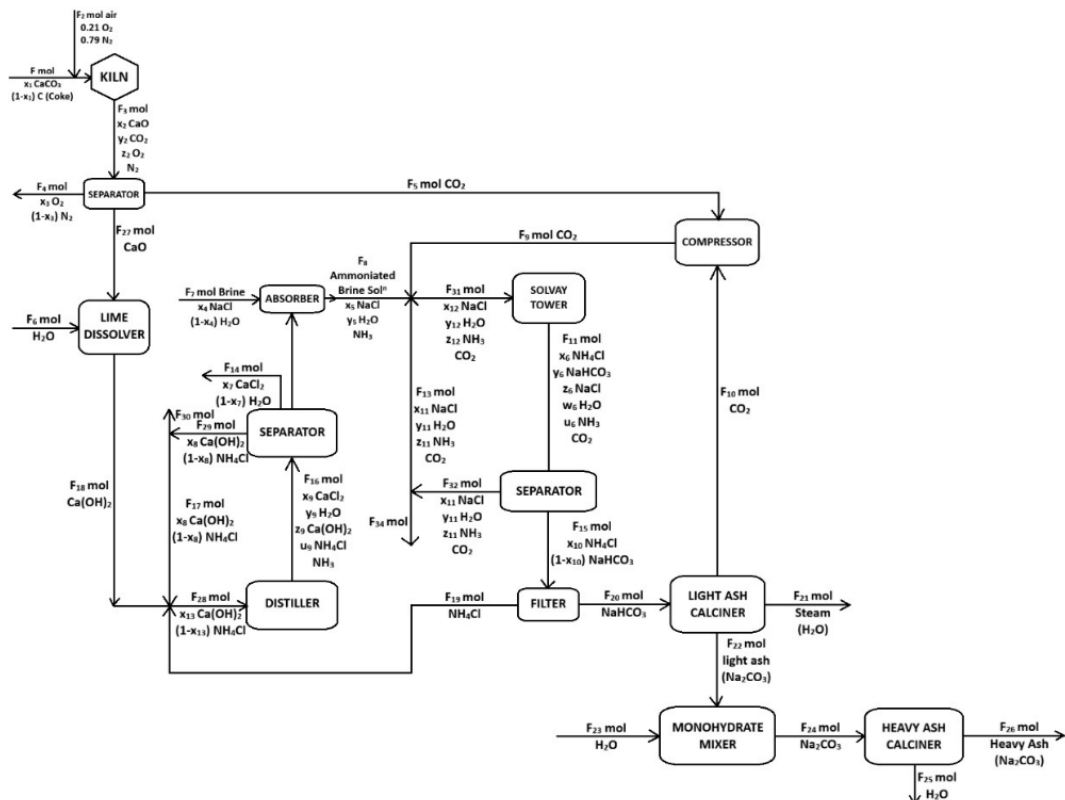
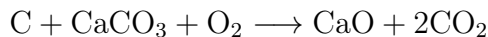


Figure 3.1: Fully Labeled Process Flow Diagram for the Solvay Process.

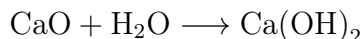
Note: All the streams are mentioned with respective flow rates (F , F_i) and component mole fractions (x_i , y_i , z_i).

3.2 Key Solvay Process Reactions

- **Kiln (Overall):**



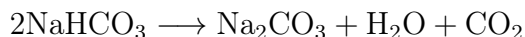
- **Lime Dissolver:**



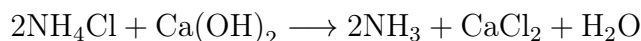
- **Carbonation Tower/Solvay Tower (Overall):**



- **Light Ash Calciner:**



- **Distiller (Ammonia Recovery):**



3.3 Basis and Given Conditions (Material Balance Specifications)

1. **Basis:** 1000 kmol/h of final dry Na_2CO_3 (Soda Ash) product.
2. **Raw Material Purity (Brine):** The incoming brine (S1) is 25% NaCl by mass, with the remainder as water.
3. **Kiln:** Limestone (CaCO_3) conversion to CaO and CO_2 is nearly 100%.
4. **Lime Dissolver:** H_2O is taken in stoichiometric ratio with CaO. Single pass conversion of CaO to Ca(OH)_2 is 100%.
5. **Absorber:** Brine is saturated with ammonia gas (nearly 6 – 8% NH_3 by mass). Brine is taken in required amount w.r.t. NH_3 . All of reactants are converted to products.
6. **Solvay Tower:** Single-pass conversion of NaCl is 86 – 91%.
7. **Light ash calciner:** Single-pass conversion of NaHCO_3 is 100%.
8. **Distiller:** Single-pass conversion of NH_3 is 95 – 98%.
9. Monohydrate mixer and heavy ash and Calciner has 100% single-pass conversion.

3.4 Operating Conditions and Unit Conversions (Provided Data)

Unit Operation	Temperature (T)	Pressure (P)	Single-Pass Conversion / Condition
Kiln	900 – 1100°C	1 atm	Conversion (CaCO_3): $\sim 100\%$
Lime Dissolver	90 – 100°C	1 atm	Conversion (CaO): $\sim 100\%$
Absorber	20 – 35°C	1 atm	Brine: $\sim 6 - 8\%$ NH_3 by mass
Solvay Tower	20 – 55°C	1 atm	Conversion (NaCl): 86 – 91%
Light Ash Calciner	160 – 230°C ($\sim 200^\circ\text{C}$)	1 atm	Conversion (NaHCO_3): $\sim 100\%$
Distiller	90 – 110°C	1 atm	Conversion (NH_3 Recovery): 95 – 98%

Chapter 4

Conclusion and Next Steps

This mid-semester report successfully defined all process flow diagrams, streams, and critical specifications, including detailed operating conditions and conversions, required to undertake the material and energy balance calculations for both the Solvay and Leblanc processes. The next steps will involve executing the hand calculations and then replicating the models within DWSIM, paying special attention to the VLE-governed separation units identified in the objectives to accurately quantify utility usage and environmental impact for the final comparative analysis.

Chapter 5

References

The following external sources were consulted during the preparation of the process flow diagrams and the material and energy balance specifications:

Leblanc Process Resources

- Wikipedia: Leblanc Process
- Academia: VARIOUS CHEMICAL PROCESSES - PRODUCTION OF SODA ASH (1. Le Blanc Process)
- Slideshare: Sodium Carbonate Industries

Solvay Process Resources

- Wikipedia: Solvay Process
- Slideshare: Soda Ash Manufacturing and Process Flow Diagram
- Scribd: Manufacturing of Soda Ash via Solvay Process
- YouTube: Solvay Process Overview (Video)